

Emotionally Intelligent Companion Dog Robot with AI-Driven Security and Voice Interaction

Sub Component (Autonomous Navigation and Obstacle Avoidance)

R26-SE-040

Project Proposal Report

SSY Wickramasinghe – IT21816086

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering (SE)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
S S Y Wickramasinghe	IT21816086	

Table 1

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof Pradeep Abeygunawardhana	14 th March 2026	

Table 2

ABSTRACT (Maximum 300 words)

The project addresses the issue of loneliness by providing an interactive robotic companion while also improving the limitations of traditional home security systems that lack autonomous mobility. Most existing robotic companion systems and security robots are expensive and rely on proprietary navigation technologies that are not easily accessible in local markets. Therefore, this research proposes an affordable robotic solution called the “Buddy” robot dog that integrates intelligent navigation and human interaction capabilities.

This individual research component focuses on developing an **autonomous navigation and obstacle avoidance subsystem** that enables the robot to move safely and intelligently within indoor environments while following a human user. The subsystem allows the robot to detect a target user, follow the user smoothly, and avoid obstacles in a realistic indoor environment.

To ensure safety and reliability, the subsystem is developed in two stages: simulation development and hardware implementation. In the simulation stage, a Python-based indoor environment is created to represent a living room containing furniture and multiple moving human agents. The robot uses a **Field of View (FOV) perception model** to detect humans and obstacles within its viewing range. A **Proportional-Derivative (PD) controller** is implemented to ensure smooth movement and stable tracking of the target human.

In the hardware deployment stage, the simulated perception system will be replaced with real sensors including a Raspberry Pi camera and ultrasonic or LiDAR sensors. The navigation algorithms will be deployed on the physical robot platform using **ROS (Robot Operating System)**.

By using affordable hardware and open-source robotics frameworks, the proposed solution aims to contribute toward **affordable intelligent robotics**, supporting Sustainable Development Goals related to innovation (SDG 9) and well-being (SDG 3).

Keywords: Autonomous Navigation, Field of View Perception, PD Controller, Obstacle Avoidance, ROS, Human-Following Robot

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, **Prof. Pradeep Abeygunawardana**, for his valuable guidance, encouragement, and support throughout the development of this research proposal.

I would also like to thank my project group members, **Ashandth Uthayashankar** and **Sobiya Anton Suresh**, for their collaboration in designing and developing the integrated Buddy Robot system.

Finally, I would like to thank the **Sri Lanka Institute of Information Technology (SLIIT)** for providing the academic resources, facilities, and support required to undertake this final year research project.

TABLE OF CONTENT (Auto Generated)

Contents

DECLARATION	2
ABSTRACT (Maximum 300 words)	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT (Auto Generated)	5
LIST OF TABLES	6
LIST OF FIGURES	7
LIST OF ABBREVIATIONS	8
LIST OF APPENDICES	8
1. INTRODUCTION	9
1.1 Background and Context.....	9
1.2 Literature Review.....	9
1.3 Research Gap	10
2. OBJECTIVE	11
2.1 Main Objective.....	11
To design and implement an autonomous navigation and obstacle avoidance system that enables the Buddy robot dog to safely navigate indoor environments while tracking and following a human user.	
2.2 Specific Objectives	11
3. METHODOLOGY	12
4. HIGH-LEVEL SYSTEM ARCHITECTURE	13
5. USER REQUIREMENTS	14
6. COMMERCIALIZATION PLAN	15
7. BUDGET AND JUSTIFICATION	16
8. WORK BREAKDOWN STRUCTURE (WBS)	17
9. GANTT CHART	18
10. REFERENCES LIST	19
11. APPENDICES	20

LIST OF TABLES

- Table 1: Student Declaration
- Table 2: Supervisor Declaration
- Table 3: Estimated Cost of Building the Robot
- Table 4: Project Timeline (Gantt Chart Phases)

LIST OF FIGURES

- Figure 1 Hybrid Control System Architecture
- Figure 2: Work Breakdown Structure

LIST OF ABBREVIATIONS

- **FOV:** Field of View
- **LiDAR:** Light Detection and Ranging
- **PD / PID:** Proportional-Derivative / Proportional-Integral-Derivative
- **ROS:** Robot Operating System
- **SDG:** Sustainable Development Goals
- **WBS:** Work Breakdown Structure

LIST OF APPENDICES

- Appendix A: PD Controller Tuning Algorithms
- Appendix B: Visual Servoing Search Pattern Logic
- Appendix C: Subsumption Architecture Priority Mapping

1. INTRODUCTION

1.1 Background and Context

With the rapid development of smart home technologies and assistive robotics, there is a growing demand for intelligent robotic systems capable of interacting with humans while performing useful tasks within indoor environments. Many individuals, particularly elderly people and individuals living alone, experience loneliness and require both emotional companionship and basic monitoring support. Robotic companion systems have emerged as a promising solution to address these needs.

Traditional home security systems rely heavily on stationary cameras and sensors that passively record activity. While these systems provide monitoring capabilities, they lack mobility and cannot actively interact with users or navigate around the environment. On the other hand, advanced robotic platforms such as robotic dogs are capable of autonomous navigation and interaction but are typically very expensive and rely on proprietary technologies.

This project aims to address these limitations by developing an affordable robotic companion system known as the **Buddy Robot Dog**. The system integrates emotional interaction capabilities together with intelligent navigation and monitoring functions.

The individual focus of this research is the development of an **autonomous navigation and obstacle avoidance subsystem**. This subsystem allows the robot to safely navigate within indoor environments while tracking and following a human user. Before implementing the system on physical hardware, the navigation logic is first developed and tested within a simulated indoor environment.

1.2 Literature Review

Autonomous mobile robots have been widely studied in the fields of robotics and artificial intelligence. Many modern robotic navigation systems rely on advanced sensor technologies such as LiDAR, depth cameras, and simultaneous localization and mapping (SLAM) algorithms to perceive their environment and navigate safely. Research in human-following robots has explored multiple approaches including vision-based tracking, sensor fusion, and machine learning-based detection methods. Visual servoing techniques using cameras are commonly used to track human movement and adjust robot motion accordingly, often utilizing Proportional-Integral-Derivative (PID) controllers to maintain smooth and stable movement.

However, many commercial robotic systems rely on expensive hardware components and proprietary navigation frameworks, making them inaccessible for low-cost consumer robotics solutions. Furthermore, ensuring smooth navigation while simultaneously avoiding obstacles in dynamic indoor environments remains a challenging problem. Robots may experience unstable motion or oscillations when reacting to moving obstacles or sudden changes in target position.

To handle complex dynamic environments, hierarchical control systems like the Subsumption Architecture proposed by Brooks [1] have proven effective by prioritizing reflexive safety behaviors over higher-level cognitive tasks. Furthermore, while traditional controllers excel at smooth target tracking, they often struggle with the non-linear nature of obstacle avoidance. Consequently, researchers have successfully implemented Fuzzy Logic

Controllers to navigate unstructured spaces, allowing robots to mimic fluid obstacle avoidance rather than rigid stop-and-turn movements. Simulating such navigation systems before hardware implementation allows researchers to evaluate system stability, safety, and reliability in controlled environments

Therefore, combining reactive obstacle avoidance strategies with stable control mechanisms such as PD controllers can significantly improve navigation performance. Simulating such navigation systems before hardware implementation also allows researchers to evaluate system stability, safety, and reliability in controlled environments.

1.3 Research Gap

Despite significant advancements in robotics, several challenges remain in the development of affordable indoor navigation systems for companion robots. Many existing robotic navigation solutions rely on expensive sensors and complex algorithms that are not suitable for low-cost robotic platforms. In addition, maintaining stable human-following behavior while simultaneously avoiding obstacles within cluttered indoor environments remains a difficult task.

Another major challenge is the limited perception capability of real-world sensors. Cameras and depth sensors typically have restricted viewing angles and ranges, meaning robots cannot perceive the entire environment at all times.

This research addresses these challenges by implementing a **Hybrid Control System (Fuzzy Logic + PID) within a priority-based Subsumption Architecture**. By decoupling "Reflexive" safety mechanisms (ESP32) from "Cognitive" navigation tasks (Raspberry Pi), the proposed system ensures robust, collision-free operation even when simulating real-world Field of View (FOV) limitations. Unlike standard stop-and-turn robots, this approach allows the robot to survive in unstructured dynamic environments by curving around obstacles smoothly, maintaining a low-cost hardware footprint while ensuring user safety.

2. OBJECTIVE

2.1 Main Objective

To design and implement an **autonomous navigation and obstacle avoidance system** that enables the Buddy robot dog to safely navigate indoor environments while tracking and following a human user.

2.2 Specific Objectives

- Simulate realistic robot behavior including human target tracking within a modeled indoor environment.
- Implement **Field of View (FOV) based perception** to replicate the sensing limitations of real robotic cameras.
- Develop a reactive obstacle avoidance system that allows the robot to navigate around obstacles smoothly.
- Implement a **Proportional-Derivative (PD) controller** to ensure stable human-following behavior without sudden or jerky movements.
- Develop an autonomous search behavior that activates when the human target moves outside the robot's field of view.

3. METHODOLOGY

The navigation subsystem is developed in two major phases: simulation development and hardware deployment.

Phase 1: Simulation Development

A Python-based simulation environment is created to represent a realistic indoor living room layout. The environment includes several obstacles such as sofas, chairs, coffee tables, and television stands. These objects simulate furniture commonly found in household environments.

Two human agents are included in the simulation. These agents move randomly within the environment while respecting room boundary constraints. The robot selects one human as the primary target and continuously adjusts its movement to follow the target.

To ensure smooth motion, a **Proportional-Derivative (PD) controller** is used to regulate the robot's velocity and direction. The controller parameters used in the simulation include:

$K_p_distance = 1.2$

$K_d_distance = 0.3$

$max_speed = 200$

These parameters allow the robot to accelerate smoothly, reduce sudden direction changes, and maintain stable tracking of the human target.

The robot perceives its environment using a **Field of View (FOV) model**, which restricts detection to a limited viewing cone defined by:

$robot_fov_angle$

$robot_fov_range$

This approach replicates the perception limitations of real sensors such as RGB cameras and depth sensors.

Obstacle avoidance is implemented using a **repulsive force behavior**, where the robot adjusts its steering direction when obstacles are detected within its viewing range. This allows the robot to navigate around obstacles while continuing to follow the human target.

Phase 2: Hardware Deployment

After validating the navigation algorithms in the simulation environment, the system will be deployed on a physical robot platform.

The simulated perception system will be replaced with real sensors including:

- Raspberry Pi camera for human detection
- Ultrasonic or LiDAR sensors for obstacle detection

The navigation algorithms will be implemented using **Python and ROS (Robot Operating System)** to enable communication between sensors, control modules, and motor actuators.

Real-world testing will be conducted in indoor environments to evaluate:

- Human following accuracy
- Obstacle avoidance reliability
- Navigation stability

4. HIGH-LEVEL SYSTEM ARCHITECTURE

System Flow:

Sensor Input → Control Processing → Subsumption Arbitration → Motor Actuation

Sensor inputs include camera data and distance sensor measurements. These inputs are processed by the navigation control system, which combines human tracking logic with obstacle avoidance behavior.

The system uses a **subsumption architecture**, where safety behaviors such as obstacle avoidance always take priority over higher-level behaviors such as human tracking. This ensures that the robot never collides with objects even when actively following a human target.

The PD controller regulates the robot's speed and direction while the obstacle avoidance system modifies the movement trajectory when obstacles are detected.

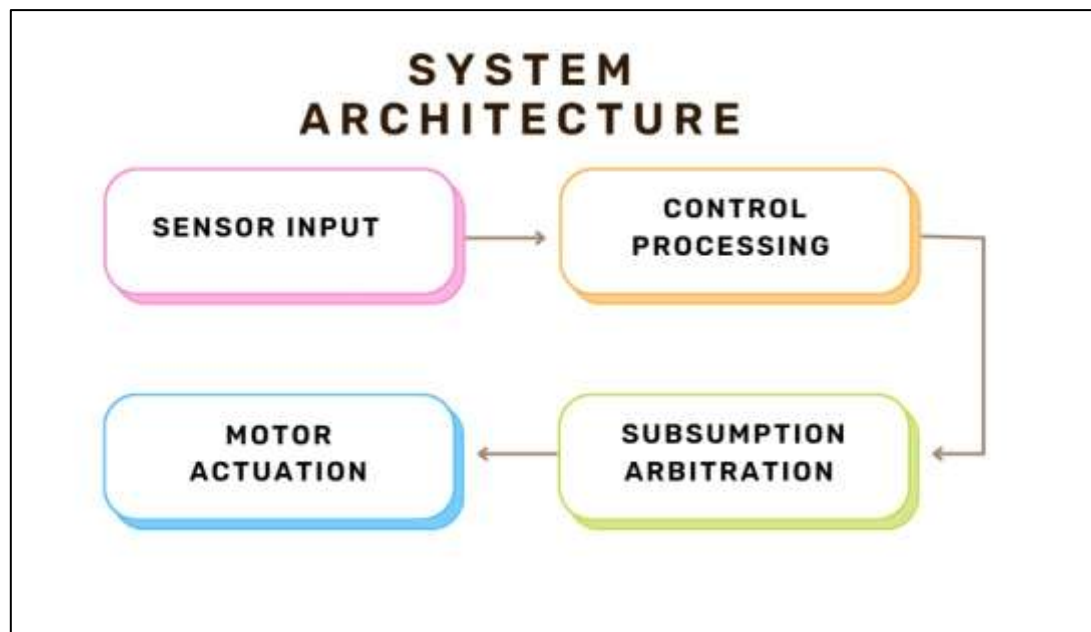


Figure 1

5. USER REQUIREMENTS

Stakeholders

- Smart home users
- Elderly individuals requiring monitoring support
- Private security companies

Functional Requirements

- The system must detect and track a human target within the robot's field of view.
- The system must autonomously avoid static and dynamic obstacles.
- The robot must smoothly follow a designated human target using PD-controlled motion.
- The system must initiate a search behavior if the human target leaves the robot's viewing range.

Non-Functional Requirements

Safety:

Collision avoidance must always take priority to prevent accidents.

Reliability:

Navigation algorithms must demonstrate stable performance in simulation before hardware deployment.

6. COMMERCIALIZATION PLAN

The Buddy robot dog has potential applications in smart homes, security monitoring, and emotional companion robotics.

Target Market

- Smart home users
- Elderly care facilities
- Small security companies
- Educational robotics institutions

Revenue Model

Revenue can be generated through:

- Hardware sales of the robot platform
- Software upgrades and feature extensions
- Mobile application services and subscriptions

Competitive Advantage

Unlike traditional home security systems, the Buddy robot integrates:

- Intelligent navigation
- Human-following capability
- Emotional interaction features

within a single affordable robotic platform.

7. BUDGET AND JUSTIFICATION

The Buddy Dog Robot prototype is developed with an approximate budget of LKR 60,000.

Component	Estimated Cost (LKR)	Justification
Raspberry Pi 4	18,000	Required for edge computing and running offline AI models.
Arduino Uno	5,000	Microcontroller for sensor integration.
MG996R Servo Motor (8–10 motors)	20,000	Actuation for the robot's limbs.
HC-SR04 Ultrasonic Sensor	1,500	Required for distance calculation and obstacle avoidance.
MPU6050 Sensor	1,200	Required for balance and movement control.
Raspberry Pi Camera Module	5,000	Required for visual input and face recognition integration.
PCA9685 Servo Driver	2,500	Required to drive multiple servo motors simultaneously.
Battery and Power System	4,000	Power supply for mobile autonomy.
Frame and Mechanical Parts	2,800	Chassis and physical structure.
Total	~60,000	

Table 3

8. WORK BREAKDOWN STRUCTURE (WBS)

- **Simulation Phase:** Develop human agents, boundary logic, living room environment, and FOV perception cone.
- **Control Logic:** Implement PD controller for tracking and repulsive force logic for smooth obstacle avoidance.
- **Hardware Acquisition:** Source the mobile robot base, computing units, sensors, and battery system.
- **Sensor Integration & Deployment:** Replace simulation perception with real camera/depth sensors and port code via ROS/Python.
- **Real-World Testing:** Evaluate human following accuracy, obstacle avoidance reliability, and navigation stability in indoor environments.

9. GANTT CHART

Phase	Duration
Research and system design	Month 1–2
AI model development	Month 3–4
Hardware integration	Month 5–6
Testing and optimization	Month 7–8
Final evaluation and documentation	Month 9

Table 4



Figure 2

10. REFERENCES LIST

- [1] R. Brooks, “A robust layered control system for a mobile robot,” *IEEE Journal on Robotics and Automation*, vol. 2, no. 1, pp. 14–23, 1986.
- [2] J. Borenstein and Y. Koren, “The vector field histogram—Fast obstacle avoidance for mobile robots,” *IEEE Transactions on Robotics and Automation*, vol. 7, no. 3, pp. 278–288, 1991.
- [3] D. Fox, W. Burgard, and S. Thrun, “The dynamic window approach to collision avoidance,” *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [4] M. Quigley et al., “ROS: An open-source Robot Operating System,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshop on Open Source Software*, Kobe, Japan, 2009.
- [5] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed. Cambridge, MA, USA: MIT Press, 2011.
- [6] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*, 2nd ed. Cham, Switzerland: Springer, 2016.
- [7] N. S. Pai, H. H. Hsieh, and Y. C. Lai, “Implementation of obstacle-avoidance control for an autonomous omni-directional mobile robot based on extension theory,” *Sensors*, vol. 12, no. 10, pp. 13947–13963, 2012.
- [8] C. Yang and K. Song, “Control design for robotic human-following and obstacle avoidance using an RGB-D camera,” in *Proceedings of the International Conference on Control, Automation and Systems*, 2019, pp. 934–939.
- [9] X. Cheng, S. Zhang, S. Cheng, Q. Xia, and J. Zhang, “Path-following and obstacle avoidance control of nonholonomic wheeled mobile robots based on deep reinforcement learning,” *Applied Sciences*, vol. 12, no. 14, 2022.
- [10] J. Peng et al., “A compliant robot control based on extended social-force model for human-following and obstacle avoidance,” *Journal of System Simulation*, vol. 35, no. 8, pp. 1776–1787, 2023.
- [11] A. Soetedjo, M. I. Ashari, and C. E. Septian, “Implementation of fuzzy logic controller for wall-following and obstacle avoiding robot,” *Journal of Applied Intelligent Systems*, vol. 4, no. 1, pp. 1–8, 2019.
- [12] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Upper Saddle River, NJ, USA: Pearson, 2004.

11.APPENDICES

- Appendix A: Simulation Environment Screenshots
- Appendix B: PD Controller Parameter Configuration
- Appendix C: ROS Node Communication Architecture
- Appendix D: Simulation Parameter Configuration